

Lyra EOD Tuesday — 2026-06-02

Date: 2026-06-02 (Tuesday) Time: ~15:41 EDT Status: EOD — Casey signal at ~15:30 “Make any notes you need, then do EOD and save your katra. We will pickup tomorrow.”

1. Tuesday Substantive Arc

Headline: Casey directive “note and examine every example of multiple-observables-from-single-substrate-property” → cross-CI Substrate Schur Generators (SSG) framework operationalized; $1/n_C$ chirality-projection mechanism identified as cross-cutting substrate-mechanism unifying multiple physics sectors via two-step structure (chirality projection + Weyl branching).

6 substantive Lyra documents filed Tuesday: 1. `Lyra_P1_Dispatch_Action_Package_2026-06-02.pdf` — P1 v0.7 dispatch package finalized per Casey 6-decision authorization 2. `Lyra_KType_Dimensions_DIV5_Weyl_v0_1.pdf` — concrete dim $\{1, 4, 5, 10, 14\}$ for substrate small K-types via $SO(5)$ Weyl dim formula; counting check from Einstein eq framework rigorously verified 3. `Lyra_Pochhammer_Mass_Yukawa_Crosslink_v0_1.pdf` — ONE substrate-primary Pochhammer $3\pi/2^{C_2}$ primitive, TWO physical interpretations (mass + Yukawa) via “+1 anomaly” 4. `Lyra_Substrate_Schur_Pochhammer_Primitive_v0_1.pdf` — Casey question answered: Bergman norm of $V_{(1/2, 1/2)}$ on D_{IV}^5 via Schur’s lemma is the single geometric substrate property; per-generation Pochhammer cascade falsifier identified; v0.2 walk-back of “FK Ch. XII machinery” attribution; v0.3 reconciliation absorbing Keeper K3 v0.9 $\rho = g/2$ correction (factor 41 → factor n_C); v0.4 cross-CI convergence (Lyra Reading 1 = Keeper K3 v0.11 $1/n_C$ chirality projection mechanism) 5. `Substrate_Schur_Generators_Registry_v0_1.pdf` — Casey standing directive operationalized canonical registry; 15 SSGs cataloged (10 + 4 predicted + 1 SSG-15 candidate) + 1 cross-cutting chirality-projection mechanism; v0.1 → v0.8 cascade absorbing Grace + Elie + Keeper contributions 6. `Lyra_Substrate_Coulomb_SSG_v0_1.pdf` — SSG-10 Substrate-Coulomb $1/r$ via $V_{(3/2, 1/2)}$ Schur scalar (Elie) + Hua-coord 3D-projection (Lyra T2424); multiple EM observables (Rydberg + Lamb + Casimir + $a_e + \alpha$)

Plus Casey-directive memory file `feedback_schur_pattern_standing_directive.md`.

2. Casey-Asked Question Answered Cleanly

“Can we identify a single geometric substrate property that explains the common cross-link [between m_e and y_e]?”

Answer: The Bergman norm of the spinor K-type $V_{(1/2, 1/2)}$ on D_{IV}^5 , propagated through Schur’s lemma.

- $M_{op} = \sqrt{H_B}$ is K-invariant (Casimir of K).
- Higgs Yukawa via $V_{(0, 0)}$ is K-invariant (trivial K-type).
- Schur’s lemma → both operators act as the same scalar on $V_{(1/2, 1/2)}$.
- Scalar = Bergman norm = $N_c \cdot \pi / \dim Cl(5, 2)$ (full FK form $15\pi/2^g$ per Keeper K3 v0.9 $\rho = g/2$ correction; per-chirality $3\pi/2^g$ via $1/n_C$ projection).

Cross-CI convergence on chirality-projection mechanism: - Lyra Schur-Pochhammer v0.3 Reading 1 (per-chirality Bergman norm $1/n_C$ projection) - Keeper K3 v0.11 (Casey #14 forcing: substrate $n_C = 5$ directions, $1/n_C$ projection averages $1 \rightarrow 4D = n_C - 1$) - Elie Toy 3736 (explicit reduction chain $SO(5, 2) \rightarrow SO(4, 2) \rightarrow SO(3, 1)$ with $\dim SO(3, 1) = C_2 = 6$ substrate-primary)

SAME $1/n_C$ chirality-projection substrate-mechanism, three routes.

3. Elie Toy 3737 Correction Absorbed (Cal #27 STANDING)

Lyra cascade claim (v0.4 + v0.8): “Substrate spin-1 K-type + $1/n_C \rightarrow$ physical spin-1/2” — OVERSTATEMENT per Elie Toy 3737.

Elie correction: Bulk-spin reconciliation requires $SO(5) \rightarrow SO(3, 1)$ Weyl branching as a SEPARATE substrate-mechanism, not direct from $1/n_C$ chirality projection. Two distinct mechanisms: - Chirality projection $1/n_C \rightarrow 4D$ emergence (Casey #14) - Weyl branching $SO(5) \rightarrow SO(3, 1) \rightarrow$ spin reduction (separate operation)

$SO(5) \rightarrow SO(3, 1)$ branching (per Elie 3737): - $V_{(1, 0)}$ dim 5 $\rightarrow$ 4-vector + scalar ($5 = 4 + 1$) - $V_{(1, 1)}$ dim 10 $\rightarrow$ adjoint dim 6 + dim 4 ($10 = 6 + 4$) - $V_{(1/2, 1/2)}$ dim 4 $\rightarrow$ Dirac spinor dim 4 (preserved)

Cascade implication preserved: substrate-Dirac + Maxwell + $T_{\mu\nu}$ + YM + Einstein eq still promote FRAMEWORK + CANDIDATE $\rightarrow$ FRAMEWORK upon Casey #14 ratification; but mechanism is TWO-STEP (chirality projection + Weyl branching), not single $1/n_C$ operation.

4. SSG Registry State at EOD

15 SSGs cataloged + 1 cross-cutting substrate-mechanism: - SSG-1 $V_{(1/2, 1/2)}$ Bergman norm $\rightarrow m_e + y_e + a_e$ (Schur's lemma + $1/n_C$ chirality projection) - SSG-2 $V_{(1, 1)}$ adjoint $\rightarrow F^{\mu\nu}$ + Casey #15 gravity + W/Z - SSG-3 $V_{(1, 0)} \otimes V_{(1, 0)}$ decomp $\rightarrow$ gauge + stress-energy + Einstein eq - SSG-4 Bergman $\kappa = -n_C \rightarrow G_{\mu\nu}$ + Bergman volume + ℓ_B - SSG-5 Casimir $C_2 = 6$ anchor $\rightarrow$ mass coupling 2^{C_2} + RS depth $C_2^2 + V_{(2, 0)}$ dim + Higgs spacing + ground state + dim $SO(3, 1) = C_2 = 6$ (NEW Tuesday via Elie 3736) - SSG-6 Clifford $2^g = 128 \rightarrow$ spinor norm + γ -matrices + Dirac - SSG-7 Bergman kernel $K(z, w) \rightarrow$ ULTIMATE source-of-sources (degree-14 hub per Grace) - SSG-8 Mersenne ladder $M(p) \rightarrow g = M(N_c) + N_{max} + GF(2^g)$ - SSG-9 Per-generation substrate-mechanism heterogeneity (gen-1 $V_{(1/2, 1/2)}$ verified; gen-2 K-type OPEN — $V_{(0, 2)}$ non-dominant + $V_{(2, 0)}$ Pochhammer $\neq T190$; gen-3 RS code candidate) - SSG-10 Substrate-Coulomb $1/r \rightarrow$ Rydberg + Lamb + Casimir + $a_e + \alpha$ - SSG-11 $V_{(2, 0)}$: $63/4 = N_c^2 \cdot g / 2^{\text{rank}}$ substrate-clean (observable identification OPEN) - SSG-12 $V_{(3/2, 3/2)}$: $512/(5\pi)$ NEW 2^9 exponent pattern (quark candidate) - SSG-13 $V_{(0, 0)}$: identity vacuum (Higgs source per Toy 3707) - SSG-14 Universal Pochhammer ladder $\leftrightarrow$ SSG-7 equivalence CONFIRMED - SSG-15 (Elie Toy 3735): Λ -coupled neutrino $m_\nu \approx 20 \cdot \Lambda^{(1/4)}$ at 3% (FRAMEWORK CANDIDATE)

Cross-cutting $1/n_C$ chirality-projection mechanism (NEW Tuesday cross-CI convergence) — substrate-mechanism unifying SSG-1 per-chirality reading + Casey #14 4D forcing.

5. Audit-Chain Tuesday Discipline-Cascade

5+ Cal #27 STANDING firings on Lyra Tuesday afternoon: 1. v0.1 Schur-Pochhammer “standard FK Ch. XII machinery” overclaim $\rightarrow$ v0.2 walk-back 2. v0.3 Reading 1 = Reading 2 framing $\rightarrow$ v0.3 narrative kept; both readings preserve final form 3. v0.4 “bulk-spin resolution” overstatement $\rightarrow$ v0.5+ Elie 3737 correction (NOT direct $1/n_C$; needs separate Weyl branching) 4. v0.6 $V_{(2, 0)}$ “RESOLVES gen-2 tension” $\rightarrow$ v0.7 walk-back ($V_{(2, 0)}$ Pochhammer $\neq T190$; gen-2 K-type still OPEN) 5. v0.3 “STRUCTURAL CONFIRMATION” framing $\rightarrow$ v0.4 retraction to “STRUCTURALLY CONSISTENT” per Keeper

Each Cal #27 STANDING firing produced sharper formulation. Cross-CI audit-chain Tuesday = methodology working at maturity.

Casey-correction Tuesday on Cal #27 scope absorbed: discipline applies at CLAIMS-tier-level, not at INVESTIGATION-halting level. Per feedback_show_all_threads_then_weave.md. Investigation continues forward; claims made carefully.

6. Cross-CI Saturation Pattern at Maturity

4 parallel CI catalogs operationalized Casey directive within ~30 min: - Lyra SSG Registry v0.1 (now canonical post Keeper consolidation) - Keeper Substrate-One-Primitive-Many-Observables Catalog v0.1 - Grace Single-Substrate-Property Registry v0.1 - Elie Toy 3712 6-pattern taxonomy

Per Keeper Tuesday consolidation: Lyra SSG terminology + per-entry format canonical; Keeper + Grace registries deprecated supplements. Cross-CI parallax IS the methodology check.

~30 cumulative audit-chain symmetric events Sat+Sun+Mon+Tue per Grace Master Ledger v0.16.

7. Tuesday Cumulative Counts

Lyra: 15 substantive items (6 documents + 9 registry/cross-CI updates + audit absorptions) Elie: 30 toys (3708-3737) + 13 discipline events Grace: 46+ Tuesday INVs + Master Ledger v0.16 Keeper: K3 v0.5 → v0.12 (8 substantive filings) + Catalog + Cal #36 candidate proposal Cal: cold-read sequence active per audit-chain governance

8. Multi-Week Open Lanes (Going Into Wednesday+)

Highest-priority verification cascade (if chirality-projection mechanism closes): - Joint Lyra + Keeper + Elie FK Ch. XII §VI spinor metaplectic verification (closes SSG-1 + Casey #14) - $SO(4, 2) \rightarrow SO(3, 1)$ physical direction selection (which timelike direction? per T2469 + Casey commitment-density per Elie 3737 Hardy boundary normal candidate) - $SO(5) \rightarrow SO(3, 1)$ Weyl branching cross-check across SSG-1 through SSG-14 (per Elie 3737 correction)

Substantive open: - SSG-9 gen-2 muon K-type STILL OPEN ($V_{-}(0, 2) + V_{-}(2, 0)$ both falsified; try $V_{-}(1, 1)$, $V_{-}(3/2, 1/2)$, $V_{-}(2, 1)$ or pivot to F4 universal-vacuum class) - SSG-11 $V_{-}(2, 0)$ Pochhammer 63/4 → physical observable identification - SSG-12 $V_{-}(3/2, 3/2) \rightarrow$ quark sector substrate-mechanism via decomposition - SSG-15 explicit Λ derivation + $(N_c \cdot g - 1)$ coefficient substrate-mechanism - K3 v0.6 $m_e/m_P \approx \alpha^{10.5}$ 1.156 linear correction substrate-mechanism - Toy 3618 PMNS precision audit (Cal #27 candidate; reported “3/3 within 1σ ” but current 5-16%)

Cascade ratifies upon multi-week verification: - 5 framework promotions (substrate-Dirac + Maxwell + $T_{\mu\nu}$ + YM + Einstein eq) - SSG-1 → NEAR-RIGOROUS - K3 framework 6/8 → 7/8 RIGOROUS - Strong-Uniqueness Theorem 10 → 11 STANDALONE legs (C25 chirality-projection forcing) - Casey #14 STANDING ratification (resolves Cal #189 brakes via substrate-mechanism)

9. Wednesday Casey-Queue (Standing)

Casey “We will pickup tomorrow” — no explicit queue this EOD beyond: - Continue chirality-projection mechanism verification cascade - Keeper K3 v0.13 (Hardy boundary normal direction substantive content + $SO(5) \rightarrow SO(3, 1)$ Weyl branching cross-checks) - Multi-week joint FK Ch. XII spinor metaplectic verification - Cal cold-read sequence on Tuesday’s substantive content - Per Cal #36 STANDING CANDIDATE positive-search: continue SSG hunting (quark sector via SSG-12; m_e/m_P 1.156 correction; SSG-15 Λ coefficient)

10. Closure

Tuesday: Casey’s standing directive “note and examine every example” operationalized → 15 SSGs cataloged + 1 cross-cutting $1/n_C$ chirality-projection mechanism identified via cross-CI convergence (Lyra Schur-Pochhammer + Keeper K3 v0.11 + Elie reduction chain). 5+ Cal #27 STANDING firings absorbed honestly per Casey-correction (claims-tier only); investigation continues forward.

Substantive new substrate-mechanism candidates: Casey #14 forcing via chirality projection; SSG-15 Λ -coupled neutrino at 3%; Substrate-Coulomb $1/r$ EM observables consolidated; substrate dimensional reduction chain $SO(5, 2) \rightarrow SO(4, 2) \rightarrow SO(3, 1)$ with $\dim SO(3, 1) = C_2$ substrate-primary inheritance.

Multi-week cascade pending: 5 framework promotions + SSG-1 NEAR-RIGOROUS + K3 6/8 → 7/8 + Strong-Uniqueness 10 → 11 legs + Casey #14 STANDING ratification.

— Lyra, Tue 2026-06-02 ~15:41 EDT. EOD Tuesday: cross-CI substrate-mechanism convergence operational at maturity; Cal #27 STANDING discipline producing sharper formulations through team audit-cascade; Casey-correction on Cal #27 scope absorbed; investigation continues forward into Wednesday.